

RIVERSIDE REGIONAL CANCER CENTER

Department of Hematologic Malignancies

February 15, 2024

Dr. Robert Kim
Primary Care Associates
2100 Medical Drive
Riverside, VA 23901

Dear Dr. Kim,

I am writing to update you on our mutual patient, Charles Jackson (DOB: 07/07/1963, MRN AML013), who is under my treatment after his diagnosis of acute myeloid leukemia.

ORIGINAL PRESENTATION (September 2021):

Mr. Jackson, then 58 years old, presented to his dentist with persistent gingival bleeding during routine cleaning. Astute observation led to CBC revealing WBC 42,000 with 62% circulating blasts, hemoglobin 8.2 g/dL, and platelets 31,000. He was urgently referred to our service on 09/20/2021.

Diagnostic workup revealed:

- Bone marrow: 77% myeloblasts
- Cytogenetics: Normal karyotype (46,XY)
- Molecular: DNMT3A R882C mutation (VAF 42%), negative for NPM1, FLT3, CEBPA
- Final diagnosis: AML with mutated DNMT3A, intermediate risk by ELN 2022

TREATMENT TIMELINE:

Induction (09/29/21): Standard 7+3 (cytarabine 100mg/m² x7 days, daunorubicin 90mg/m² x3 days - higher dose given age <60). Course complicated by:

- Viridans streptococci bacteremia (day 8)
- Grade 3 mucositis requiring TPN
- Prolonged neutropenia (34 days)

Day 28 marrow: CR with MRD 0.3% by flow cytometry

Consolidation (Nov 2021 - March 2022): Received 4 cycles HiDAC. Cycle 2 required dose reduction to 2g/m² due to cerebellar toxicity (ataxia, dysmetria - fully resolved).

MRD Assessment (April 2022): Post-consolidation BM: Morphologic CR, MRD negative by flow and NGS (DNMT3A mutation undetectable, sensitivity 0.01%)

RELAPSE (May 2023):

Routine CBC on 05/18/23 showed WBC rising to 8,900 with 15% blasts. Marrow confirmed relapse with 43% blasts, DNMT3A mutation re-emerged (VAF 38%), new NRAS mutation detected.

SALVAGE THERAPY AND TRANSPLANT:

Given excellent PS and CR duration of 20 months, pursued salvage with FLAG-IDA (June 2023):

- Fludarabine 30mg/m² days 2-6
- Cytarabine 2g/m² days 2-6
- Idarubicin 10mg/m² days 4-6
- G-CSF days 1-7

Achieved CR2 (day 30 marrow: 2% blasts, MRD negative)

Allogeneic HSCT (August 2023):

- Donor: 9/10 HLA matched unrelated (single DQ mismatch), CMV matched
- Conditioning: Fludarabine/Busulfan (myeloablative)
- GVHD prophylaxis: Tacrolimus/Methotrexate
- Infused 6.2 x 10⁶ CD34+ cells/kg on 08/15/23

POST-TRANSPLANT COURSE:

Engrafted day +14 (ANC >500) and day +22 (platelets >20K)

Complications:

- Grade 2 acute skin GVHD (day +42): Responded to prednisone 1mg/kg
- CMV reactivation (day +65): Treated with valganciclovir x6 weeks
- Mild chronic GVHD of mouth (month 4): Managed with topical steroids

Current Status (February 2024, Day +180):

- Complete remission maintained
- 100% donor chimerism (checked monthly x6)
- Off immunosuppression since January 2024
- ECOG PS 1, back to work part-time as accountant

CURRENT MEDICATIONS:

1. Acyclovir 800mg BID
2. Fluconazole 200mg daily (through month 6)
3. Trimethoprim-sulfamethoxazole DS 3x/week
4. Ursodiol 300mg TID
5. Vitamin D 2000 units daily
6. Folic acid 1mg daily

RECENT STUDIES (02/01/24):

- CBC: WBC 4.8K (normal differential), Hgb 12.1, Plt 156K
- Chemistry: Cr 1.1, ALT 42, AST 38, Bili 0.9
- Bone marrow: Normocellular, no AML, 100% donor chimerism
- PFTs: Mild restrictive pattern (FEV1 78% predicted)
- Echo: Normal EF 65%

MONITORING PLAN: We continue close surveillance given intermediate-risk disease and prior relapse:

- Monthly visits through month 12
- CBC/CMP monthly
- Bone marrow biopsy q3 months year 1, q6 months year 2
- Chimerism studies with each marrow
- PFTs q3 months (watching for bronchiolitis obliterans)

PROGNOSIS: While his initial 20-month remission was encouraging, relapse indicates more aggressive disease biology despite isolated DNMT3A mutation initially. However, achieving CR2 and successful transplant with 100% donor chimerism at 6 months provides approximately 35-40% chance of long-term cure.

Please continue routine health maintenance with attention to:

- Skin cancer screening (increased risk post-transplant)
- Bone density monitoring (chronic GVHD + steroids)
- Revaccination series beginning at 12 months post-transplant (we will coordinate)
- Cardiovascular risk factors (statins held during acute treatment, consider restarting)

Thank you for your continued partnership in Mr. Jackson's care. Please call me directly at 555-8892 with any concerns.

Sincerely,

Margaret Chen, MD
Director, Blood and Marrow Transplant Program